
Microprocessor Systems and Interfacing

EEE 342

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Lecture # 09

Week 6 Lecture 2

Microprocessor 8088

■ Outline

- Interfacing with 8088

Memories

- Some Important numbers to consider
 - ❑ 1K = 1024 Location → 10 Address Pins, 2^{10}
 - ❑ 2K = 2048 Location → 11 Address Pins 2^{11}
 - ❑ 4K = 4096 Location → 12 Address Pins 2^{12}
 - ❑ 1M = 1 Mega Location → 20 Address Pins
 - ❑ 400H represents 1K-byte of Memory
 - ❑ 1000H represents 4K
 - ❑ 2000H represents 8K

Memories

■ Memory Organization

- Each memory location contains 2^X locations, where x is the number of address pins on the chip
- Each memory location contains y bits, where y is number of data pins on the chip.
- Total bits = $2^X * Y$ bits

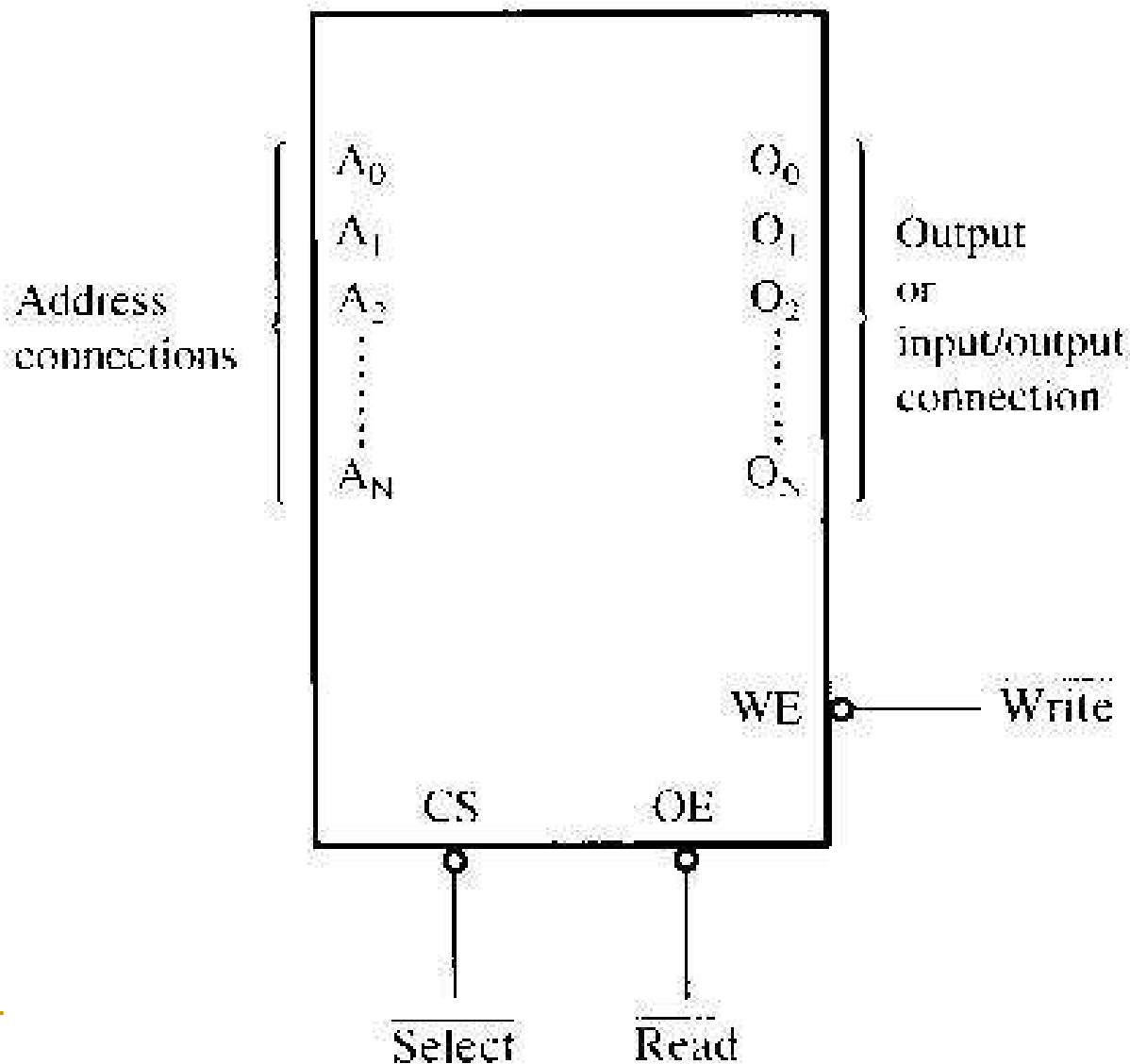
■ Memory Speed

- Access Time

Pin configuration of Memory Chips

- Pin connections common to all memory devices.
 - ❑ The address inputs
 - ❑ Data outputs or input
 - ❑ Some type of selection input
 - ❑ At least one control input used to select a read or write operation.
- Memory device has one chip select, chip enable or select
- Memory device has control pin

Pin configuration of Memory Chips



NAND Gate Decoder

- Interface 16KB SRAM with 8088 using NAND Decoder. Available chip size is 4KB. Memory map starts at address 8E000h